Section 3: Reproduction and inheritance

a) Reproduction

b) Inheritance

a) Reproduction

Students will be assessed on their ability to:

3.1 describe the differences between sexual and asexual reproduction

3.2 understand that fertilisation involves the fusion of a male and female gamete to produce a zygote that undergoes cell division and develops into an embryo

Flowering plants

3.3 describe the structures of an insect-pollinated and a wind-pollinated flower and explain how each is adapted for pollination

3.4 understand that the growth of the pollen tube followed by fertilisation leads to seed and fruit formation

3.6 recall the conditions needed for seed germination

3.7 understand how germinating seeds utilise food reserves until the seedling can carry out photosynthesis

3.8 understand that plants can reproduce asexually by natural methods (illustrated by runners) and by artificial methods (illustrated by cuttings)

Humans

3.9 recall the structure and function of the male and female reproductive systems

3.10 understand the roles of oestrogen and progesterone in the menstrual cycle

3.11 describe the role of the placenta in the nutrition of the developing embryo

3.12 understand how the developing embryo is protected by amniotic fluid

3.13 recall the roles of oestrogen and testosterone in the development of secondary sexual characteristics.

b) Inheritance

3.14 recall that the nucleus of a cell contains chromosomes on which genes are located

3.15 understand that a gene is a section of a molecule of DNA

3.16 describe a DNA molecule as two strands coiled to form a double helix, the strands being linked by a series of paired bases: adenine (A) with thymine (T), and cytosine (C) with guanine (G)

3.17 understand that genes exist in alternative forms called alleles which give rise to differences in inherited characteristics

3.18 recall the meaning of the terms: dominant, recessive, homozygous, heterozygous, phenotype, genotype and **codominance**

3.19 describe patterns of monohybrid inheritance using a genetic diagram

3.20 understand how to interpret family pedigrees

3.21 predict probabilities of outcomes from monohybrid crosses

3.22 recall that the sex of a person is controlled by one pair of chromosomes, XX in a female and XY in a male

3.23 describe the determination of the sex of offspring at fertilisation, using a genetic diagram

3.24 understand that division of a diploid cell by mitosis produces two cells which contain identical sets of chromosomes

3.25 understand that mitosis occurs during growth, repair, cloning and asexual reproduction

3.26 understand that division of a cell by meiosis produces four cells, each with half the number of chromosomes, and that this results in the formation of genetically different haploid gametes

3.27 understand that random fertilisation produces genetic variation of offspring

3.28 recall that in human cells the diploid number of chromosomes is 46 and the haploid number is 23

3.29 understand that variation within a species can be genetic, environmental, or a combination of both

3.30 recall that mutation is a rare, random change in genetic material that can be inherited

3.31 describe the process of evolution by means of natural selection

3.32 understand that many mutations are harmful but some are neutral and a few are beneficial

3.33 understand how resistance to antibiotics can increase in bacterial populations

3.34 understand that the incidence of mutations can be increased by exposure to ionising radiation (for example gamma rays, X-rays and ultraviolet rays) and some chemical mutagens (for example chemicals in tobacco).